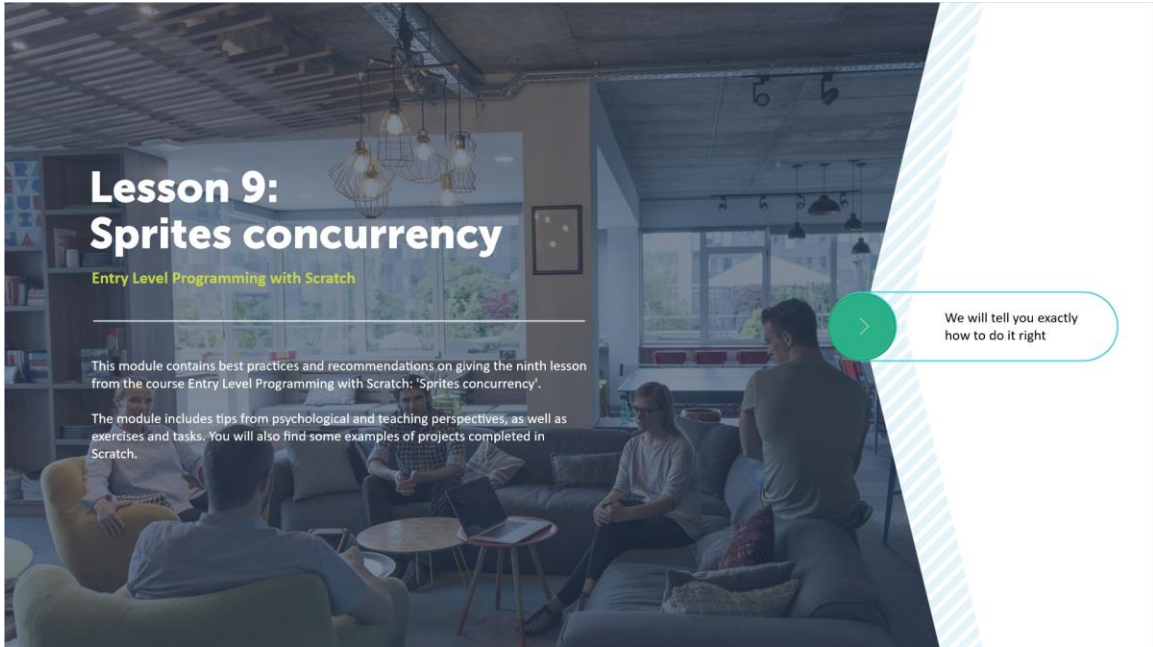




Lesson 9: Sprites concurrency

Entry Level Programming with Scratch

This module contains best practices and recommendations on giving the ninth lesson from the course Entry Level Programming with Scratch: 'Sprites concurrency'.

The module includes tips from psychological and teaching perspectives, as well as exercises and tasks. You will also find some examples of projects completed in Scratch.

We will tell you exactly how to do it right



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STAGE 1. Homework check and learned material update

Goal:

answer the questions your group might have while completing homework. Engage them in the learning process, revise the material from the previous lesson; give students a chance to present their projects in front of the group. Gradually they will learn how to touch upon the main points in brief.

Homework check

Start the lesson by **checking the homework** and **answering the questions** your students have come up with while completing home tasks. **Answering the questions** and **correcting mistakes** are the main tasks of the instructor at the beginning of the lesson. Discussing mistakes will help to prevent children from doing them again.

1

Homework check

2

Question-Answer

3

Correction of mistakes





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STAGE 1. Homework check and learned material update

During this part of the lesson, **children comment** on other projects and discuss them in Scratch Studio.

Commenting on projects and discussing them are important parts of homework.

There are two options to discuss homework:

At class
Presentation, commenting on other projects and discussion **at class** are important parts of homework which develop presentation and analytical skills.

In group chat
Commenting, formulating questions, polite discussion of other projects **in chats** are very important components of homework that develop communicative skills and the ability to work in groups.

The duration of this stage depends on the difficulty of the homework. Usually, it takes 10-20 min.

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STAGE 2. Synchronizing animated series

Goal:

introduce the concept of concurrency, use examples to explain it.

The overall logic of the explanation

Start explaining the topic of concurrency with an example **'The Turnip'**. Based on the **scripted plan** the project demonstrates its difference from the **unpredictable processes** [the processes that depend on the reaction of the programmer/computer].

When it comes to the unpredictable processes sprites continue **intercommunicating**; they may compete against each other, fight for resources, objects etc. You can organize their cooperation with the help of **synchronization**.

STAGE 2. Synchronizing animated series

Ways to synchronize objects and events in Scratch:

Method 1

With the help of
'broadcast' and
'receive' command
blocks.

Method 2

With the help of a
variable.



Try to use only those concepts and terms that
your students are already familiar with!

The examples of projects that can be used to explain different types of
synchronization:

1

'The Turnip'

2

'The Cup'

After you have analyzed the example move on to **explain the concept of
synchronization with the help of thematic projects.**

Let's take a closer look at each project (the first part of the lesson will
be dedicated to the 'The Turnip' project. After the break, we will return
to another project - 'The Cup').

STAGE 2. Synchronizing animated series

Exercise

Project 1. 'The Turnip'

[Link](#) to the project.

'The Turnip' is a ready-made project with synchronization of objects
used for animation script. This is a vivid example of the well-
coordinated operations which follow a programmed plan. Some
operations are executed simultaneously while the others follow one
another, that is comply with a sequence.

Task:

- launch the project
- say which operations are **simultaneous/successive**

After launching the project several times the students will
understand that **an animated series repeats the same process time
and again.**



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STAGE 3. Energizer-break

Goal:

to let children expend some energy and rest by changing the activity.

The energizer-break allows to combine physical activity and the learned material.








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STAGE 3. Energizer-break

Use the complicated version of the game 'Control the cat' from the Lesson 7 break.

'Control the cat':

- Children split into 3 teams: 'chief' (programmer), 'cat' (Sprite) and 'obstacles'.
- Put a small paper illustration of a cat around the neck of each Cat so that each student could associate himself or herself with the Sprite. Cats get blindfolded whilst other kids spread around the room to create obstacles.
- The task of a chief is to command a cat and lead him or her successfully through the room to the aim (for example, to a candy). In order to do this, a chief uses Scratch commands 'move 10 steps', 'turn 90 degrees' and etc.

!

The difference from the Lesson 1 break is that cats do not have one single destination. They may have several tasks to complete: for example, collect all toys in the labyrinth.






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STAGE 4. Synchronizing moving objects

Goal:

consolidate the material from Part 1 and give insights on the topic by introducing more complicated tasks.

After the break continue with explaining the messages between sprites and synchronization with the help of example projects.

How can we move from 'The Turnip' to the animated series projects?

Explain to students that programming is very similar to the real-life, as many events and objects cannot be defined beforehand. For some reason, **some things happen earlier or later**; it's impossible to predict anything; however, you need to take action.

Search for the examples in the real-life how different factors influence the course of life (weather, mood etc.).

When it comes to a computer, different programs compete for resources when following their algorithms:

- Videocamera
- Printer
- Speakers

To synchronize their operation different devices and programs exchange messages and signals: whether they're busy or ready for work.

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STAGE 4. Synchronizing moving objects

Exercises

Project 2

'The Cup'

[Link](#) to the project.

Key points: Several sprites try to make it to the goal (cup). As soon as one of the sprites touches the cup, it broadcasts the message ('i have made it'). The sprite which takes over the control owns the cup. Sprites touch the object in turns.

Try to explain that **there's no way two sprites can own one object**.

Note: This project is similar to the concept of mutex – each sprite has its own script (just like the modern programming which is based on multitasking).

Programs sync because they compete for different resources (for example, for a printer).

!

This project is an example of the object synchronization. It doesn't matter how sprites synchronize; the only thing that matters is that synchronization eliminates competition for the same resources.

Project 'The Turnip'

If you wish you can use the project 'The Turnip' and the template to complete the project.

Task for students: Try to complete the 'Turnip' script to return to the original version of the fairy-tale.

[Link](#) to the template.

Try to provide the idea that **non-determinism is more interesting** than just following a given scenario.

STAGE 4. Synchronizing moving objects

Try get the following idea across to the students: **unpredictables things are fun** and following the script is boring.

Hand out stickers for kids to glue them into their exercise books.



Stickers can be distributed at the end of each lesson to consolidate the information from the lesson. You can also hand them out at the beginning of the lesson when explaining the new material or even before the lesson - as a small cheat note.



STAGE 5. Homework and reflection

Goal:

discuss the lesson, give your students homework and explain it.

Reflection

Reflection is a good way to find out **what students have learned during the lesson** and **which things still seem complicated and need explanation**.

Use the interactive board to help students. Show them a tag cloud that will help kids to express their thoughts.

Examples:

- 'Today I've learned about...'
- 'It was difficult to...'
- 'I understood/ I didn't understand...'
- 'I have learned how to...'
- 'I managed to...'
- 'It was interesting that...'
- 'I was surprised that...'
- 'I felt like...'

Each student chooses 1-2 tags and makes up the ending.



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STAGE 5. Homework and reflection

The homework is divided into 3 levels:

Level 1

Level 2

Level 3

This is an obligatory task for all students. Children are required to do an exercise similar to the one they have done during the lesson.

!

It's important to avoid splitting students into groups by their capabilities because later there'll be no actual division.

Student with best study skills become an 'educator's assistant'. Ask them to explain the material and put it in practice together with those children who lag behind. Be careful not to make kids think that their education process is deliberately slowed down.



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STAGE 5. Homework and reflection

The homework is divided into 3 levels:

Level 1

Level 2

Level 3

An advanced task marked with an asterisk — meant for those students who have done well during the class and can dig into the subject.

!

It's important to avoid splitting students into groups by their capabilities because later there'll be no actual division.

Student with best study skills become an 'educator's assistant'. Ask them to explain the material and put it in practice together with those children who lag behind. Be careful not to make kids think that their education process is deliberately slowed down.





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STAGE 5. Homework and reflection

The homework is divided into 3 levels:

Level 1

Level 2

Level 3

This is a task for those students who **have time and abilities to complete difficult projects**.

!

It's important to avoid splitting students into groups by their capabilities because later there'll be no actual division.

Student with best study skills become an 'educator's assistant'. Ask them to explain the material and put it in practice together with those children who lag behind. Be careful not to make kids think that their education process is deliberately slowed down.







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Additional materials

Additional materials for lesson 9:

File

General rules.pdf



File

Stickers.pdf







Close Lesson

